ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Resources and Development (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** identification and inventory by surveying and mapping	Counting what exists
2	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
3	**(c)** terrace farming	Terraces slow the water
4	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
5	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
6	**(a)** laterite - alluvial deposits of rivers	Laterite is formed by leaching
7	**(b)** laterite soil	Leached of nutrients
8	**(d)** A is false but R is true.	A false; R true
9	**(a)** Both A and R are true and R is the correct explanation of A.	Cotton needs moisture-retentive soil; black (regur) soil is clayey and holds moisture, so R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true statements about two different soils; R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Soil supports plant growth and forms the basis of agriculture (1); It provides nutrients, water and a home to living things (1).	1+1
12	Technology can exploit resources faster than they renew (1); Without care it leads to depletion and pollution (1).	1+1
13	It causes waterlogging and brings salts to the surface (1); The soil becomes saline and loses fertility, as in Punjab and Haryana (1).	1+1
14	Pastures there are limited and dry (1); Large herds graze for long on the same land and strip it bare (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Overgrazing strips the vegetation (1); Over-irrigation causes salinity (1); Excess chemicals damage soil life (1).	1+1+1
16	On hills, terraces and contour ploughing check run-off (1); On plains, strip cropping and shelter belts check wind and water (1); Afforestation helps in both (1).	1+1+1
17	Mining degrades land in Jharkhand (1); Overgrazing damages land in Gujarat and Rajasthan (1); Over-irrigation causes salinity in Punjab and Haryana (1).	1+1+1
	OR: Gandhi's quote warns against greed (1); Overuse of resources today causes depletion (1); It supports sustainable use for the future (1).	1+1+1
18	India has a large population on a small land share (1); Pressure on land leads to more intensive use and degradation (1); Land-use planning is needed for food and development (1).	1+1+1
	OR: It is fertile with minerals (1); It lies in plains with rivers and irrigation (1); It supports intensive farming that feeds large populations (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Regur soil (1). (ii) Volcanic rocks (lava) (1). (iii) Any one of wheat, sugarcane, soyabean, jowar, linseed, citrus fruits or vegetables (1). (iv) In the dry season it develops deep cracks, which allow air to enter and loosen and aerate the soil, as if it were ploughed itself (2).	1+1+1+2

20	Land degradation results from mining (Jharkhand, Chhattisgarh, Madhya Pradesh, Odisha), overgrazing (Gujarat, Rajasthan, Madhya Pradesh, Maharashtra) and over-irrigation (Punjab, Haryana, western Uttar Pradesh) (1); Mining leaves deep scars after the removal of vegetation, and mineral processing creates dust that settles on the land and slows infiltration of water (1); Over-irrigation causes waterlogging that increases salinity and alkalinity of the soil (1); Industrial effluents and waste also pollute land and water (1); Remedies include afforestation, proper management of grazing, planting shelter belts, stabilising sand dunes with thorny bushes, controlling mining and treating industrial effluents before discharge (1).	1+1+1+1+1
	OR: On the basis of origin, resources are biotic (forests, animals, fisheries) and abiotic (rocks, metals, minerals), which tells us where they come from (1); On the basis of exhaustibility they are renewable (solar and wind energy, water, forests) or non-renewable (minerals and fossil fuels that take millions of years to form) (1); On the basis of ownership they are individual, community, national or international, which decides who may use and manage them (1); On the basis of status of development they are potential, developed, stock or reserves; for example wind energy in Rajasthan and Gujarat is potential, while river water in a dam is a reserve (1); The classification is practical because planners can see which resources are limited, who controls them and how far they can be used with the available technology (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up classifications: **renewable/non-renewable** (by exhaustibility) and **biotic/abiotic** (by origin) are different; coal is both biotic and non-renewable.
- Confusing **stock** (not yet usable for lack of technology) with **reserves** (the part of stock that can be put to use with existing technology).
- Swapping **bangar** (older alluvium, kankar nodules) and **khadar** (newer alluvium, more fertile).
- Writing that red soil is formed from alluvium; it develops on **crystalline igneous and metamorphic rocks**, and its colour is due to iron compounds.
- Calling black soil 'alluvial' or forgetting it is poor in phosphoric content though rich in lime, potash and magnesium.
- Dating the Earth Summit and **Agenda 21** at Rio de Janeiro wrongly; both are **1992**, not Kyoto or Paris.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Resources and Development (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** careful resource planning	Balanced development
2	**(d)** regur soil - cotton	Black soil retains moisture
3	**(c)** overgrazing	People overuse land
4	**(c)** potential resource	Not yet utilised
5	**(b)** over-exploitation	Excessive use exhausts resources
6	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
7	**(a)** Both A and R are true and R is the correct explanation of A.	Regional imbalance
8	**(c)** they can be used with the present technology but are not yet fully used	Reserves are part of stock
9	**(c)** A is true but R is false.	A is true. Bangar, the older alluvium, has a higher concentration of kankar nodules, so R is false.
10	**(d)** A is false but R is true.	Renewable resources can be replenished by physical, chemical or mechanical processes, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Local people know and use the resource daily (1); Their involvement ensures care and enforcement of rules (1).	1+1
12	It shows that resources are created by human use, not just by nature (1); Technology and culture decide what counts as a resource (1).	1+1
13	Tree roots bind the soil and canopy breaks the rain's force (1); Forests slow run-off and add humus (1).	1+1
14	Colonial powers exploited resources for their benefit (1); This left regions poor and unequal in development (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Jharkhand is mineral rich but economically backward (1); Rajasthan has solar and wind but little water (1); Arunachal Pradesh has water but lacks infrastructure (1).	1+1+1
16	The Rio Summit adopted Agenda 21 for sustainable development (1); It urged nations to protect resources (1); It asked local governments to prepare their own plans (1).	1+1+1
17	Roots hold the soil together (1); The canopy reduces rain impact (1); Leaf litter slows the flow of water (1).	1+1+1
	OR: Colonial powers extracted resources for their own use (1); Regions became dependent and underdeveloped (1); The effects still shape resource distribution (1).	1+1+1
18	India has a large population and limited resources (1); Sustainable development avoids damage to the environment (1); It balances present and future needs (1).	1+1+1
	OR: Gullies cut fields into pieces (1); Land becomes unfit for cultivation (1); Farmers lose crops and income, and ravines like those of the Chambal form (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Community-owned resources (1). (ii) The village pasture or pond (1). (iii) 12 nautical miles (1). (iv) International institutions control them; they belong to everyone, so any use needs international agreement (2).	1+1+1+2

20	Sheet erosion removes the top layer of soil over a wide area as water flows as a sheet on gentle slopes, and it often goes unnoticed (1); Gully erosion occurs when running water cuts deep channels in clayey soil, making land unfit for cultivation, as in the ravines of the Chambal (1); Wind erosion removes loose soil in dry and desert regions (1); On slopes, contour ploughing slows water, terrace farming cuts steps to reduce run-off and strip cropping with grass strips breaks water flow and wind (1); Shelter belts of trees check wind speed and have helped to stabilise sand dunes in western India, while afforestation holds soil with roots (1).	1+1+1+1+1
	OR: Mining in Jharkhand, Chhattisgarh, Madhya Pradesh and Odisha has caused deforestation and left deep scars on the land (1); Mineral processing generates dust that settles on the surface and prevents water from infiltrating the soil (1); Overgrazing in Gujarat, Rajasthan, Madhya Pradesh and Maharashtra strips the land of its vegetation cover (1); Both lead to erosion, loss of fertility and eventually land unfit for cultivation (1); Controlling mining, regulating grazing, afforestation and restoring mined land are needed to check this (1).	1+1+1+1+1

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- Confusing **stock** (not yet usable for lack of technology) with **reserves** (the part of stock that can be put to use with existing technology).
- Swapping **bangar** (older alluvium, kankar nodules) and **khadar** (newer alluvium, more fertile).
- Writing that red soil is formed from alluvium; it develops on **crystalline igneous and metamorphic rocks**, and its colour is due to iron compounds.
- Calling black soil 'alluvial' or forgetting it is poor in phosphoric content though rich in lime, potash and magnesium.
- Dating the Earth Summit and **Agenda 21** at Rio de Janeiro wrongly; both are **1992**, not Kyoto or Paris.

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MASTERM

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Resources and Development (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** water run-off and soil loss	Contour ploughing
2	**(c)** the Rio Earth Summit of 1992	It adopted Agenda 21
3	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
4	**(a)** stock	Needs appropriate technology
5	**(d)** it cannot be used because the technology is not available	Example: water as hydrogen and oxygen
6	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
7	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
8	**(b)** black soil	Regur retains moisture
9	**(a)** Both A and R are true and R is the correct explanation of A.	Slowing the runoff reduces soil erosion; R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R describes Agenda 21 rather than explaining the date or venue of the Summit.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Planning matches resources to local needs (1); It invests in backward regions and allocates technology, capital and infrastructure (1).	1+1
12	Renewable resources like forests and water can be used up faster than they regenerate (1); Overuse and pollution can make them scarce (1).	1+1
13	Sheet erosion removes a thin layer of soil over a large area (1); The fertile top layer is lost slowly and farmers do not notice it (1).	1+1
14	Resources may lack infrastructure, capital and technology (1); Colonial-era exploitation and unequal use kept benefits away from local people (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Communities own and manage commons like pastures (1); They enforce rules for sustainable use (1); Local involvement makes conservation effective (1).	1+1+1
16	Renewable resources can be exhausted if overused (1); Forests and water can be degraded (1); Conservation ensures that they regenerate for the future (1).	1+1+1
17	Technology can create new resources, like hydroelectric power (1); It can also deplete them by over-exploitation (1); Careful use is therefore essential (1).	1+1+1
	OR: Alluvial soil suits wheat, rice and sugarcane (1); Black soil suits cotton (1); Laterite suits tea, coffee and cashew (1).	1+1+1
18	Resources do not exist by themselves but become useful with human technology (1); Culture and economy decide what is valued (1); Hence people play a key role in resource creation (1).	1+1+1
	OR: Contour bunds slow water on slopes (1); They reduce soil loss in heavy rainfall (1); They help farmers retain moisture (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	(i) Gully erosion (1). (ii) Bad land (in Chambal basin, ravines) (1). (iii) Terrace farming (1). (iv) In strip cropping, strips of grass are left between crops to break the force of wind and water; shelter belts of trees check wind and stabilise the soil (2).	1+1+1+2
20	India has about 2.4 per cent of the world's land area but supports about 16 per cent of the world's population, so the pressure on land is very high (1); Land is needed at once for farming, forests, pastures, industry, roads and settlements, so land use must be planned (1); Net sown area, forests, fallow land, barren and wasteland and land under non-agricultural uses together describe how land is used (1); Degradation by overgrazing, mining and over-irrigation reduces productivity at the very time food demand is rising (1); Economic development therefore depends on sustainable land use, conservation of soil and balanced allocation of land among competing uses (1).	1+1+1+1+1
	OR: Soil is the medium of plant growth and supports most of the farming on which Indians depend (1); It is a living system with organisms, humus, minerals and water, and takes thousands of years to form (1); Different soils like alluvial, black and red decide which crops can be grown in a region (1); Erosion and degradation reduce fertility faster than nature can restore it (1); Soil conservation by contour ploughing, terracing, strip cropping and shelter belts is therefore vital for food security (1).	1+1+1+1+1

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